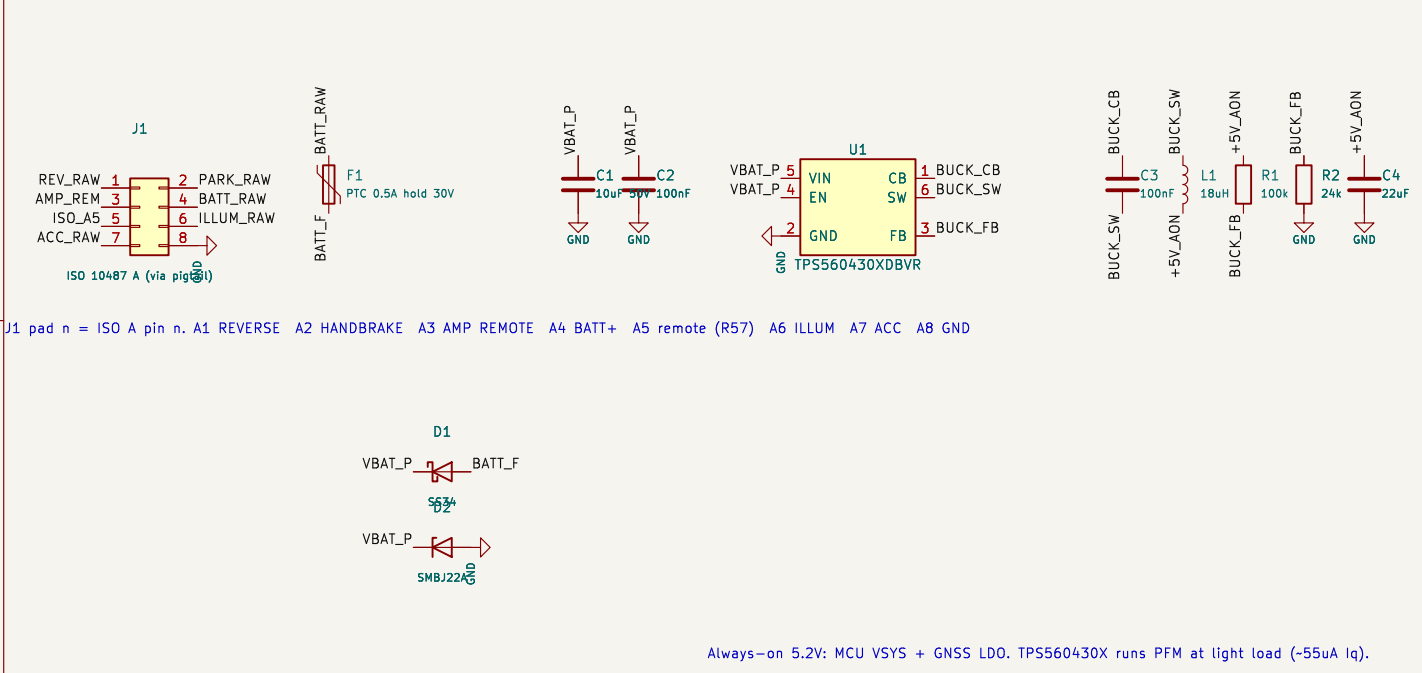


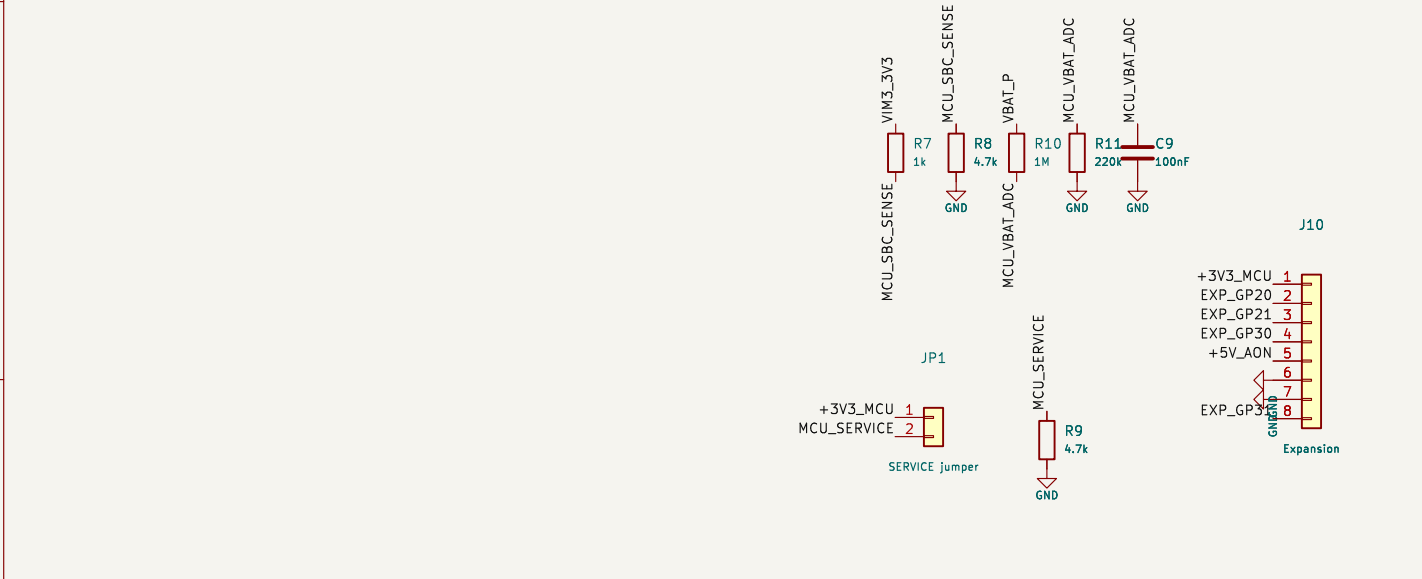
1. ISO 10487 CONNECTOR A (power) + ALWAYS-ON 5V FOR PIC0 AND GNSS



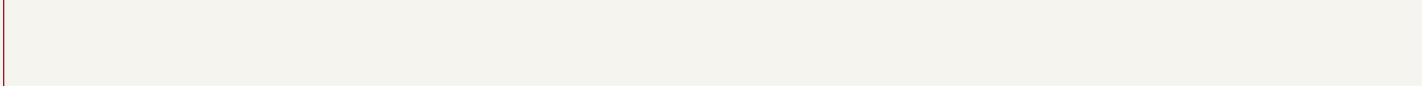
2. VEHICLE INPUTS (opto-isolated, ACC, ILLUM, REVERSE: +12V = on, PARK: switch to ground = on.



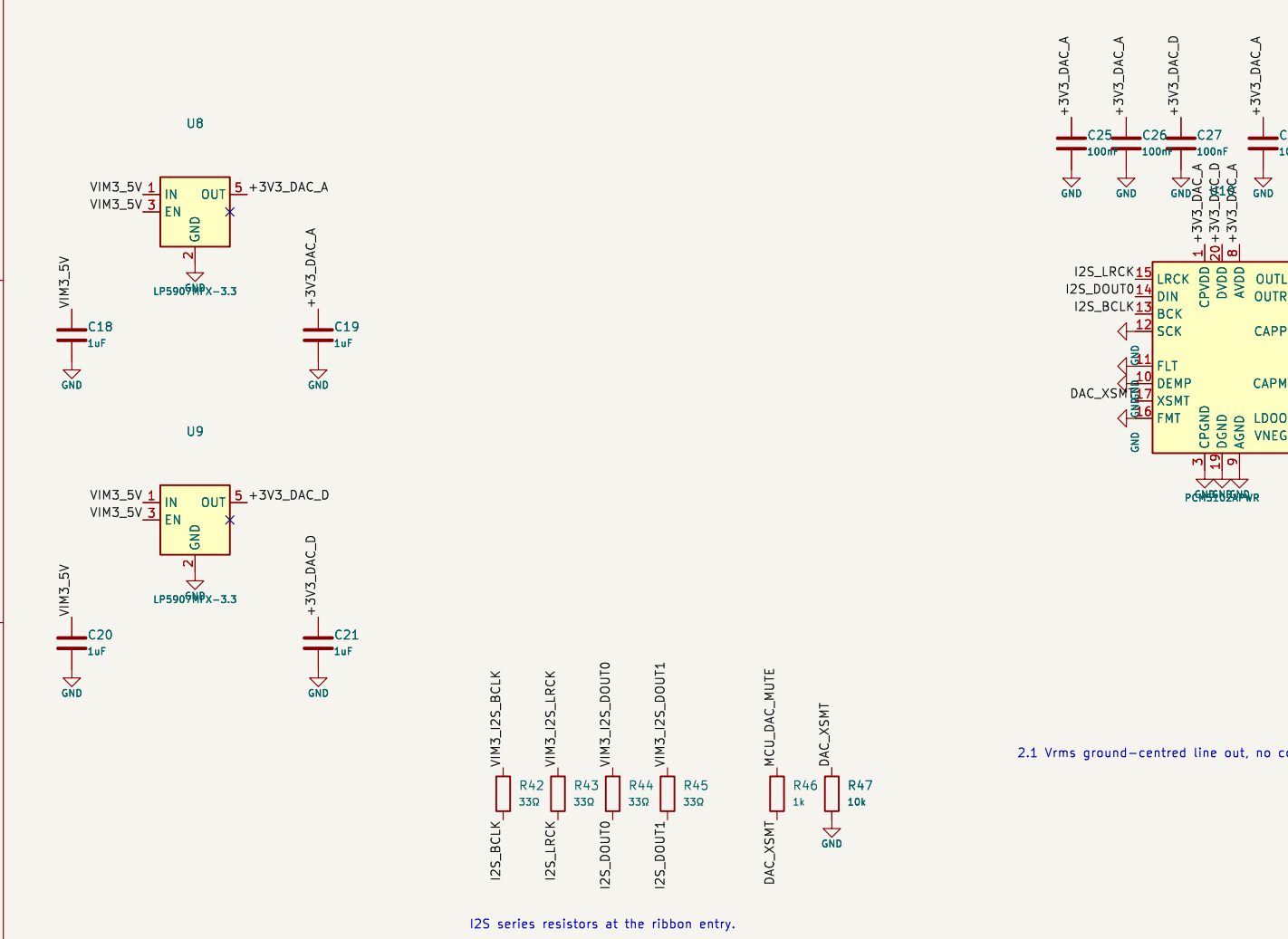
4. MCU HELPERS (MCU core is section 13)



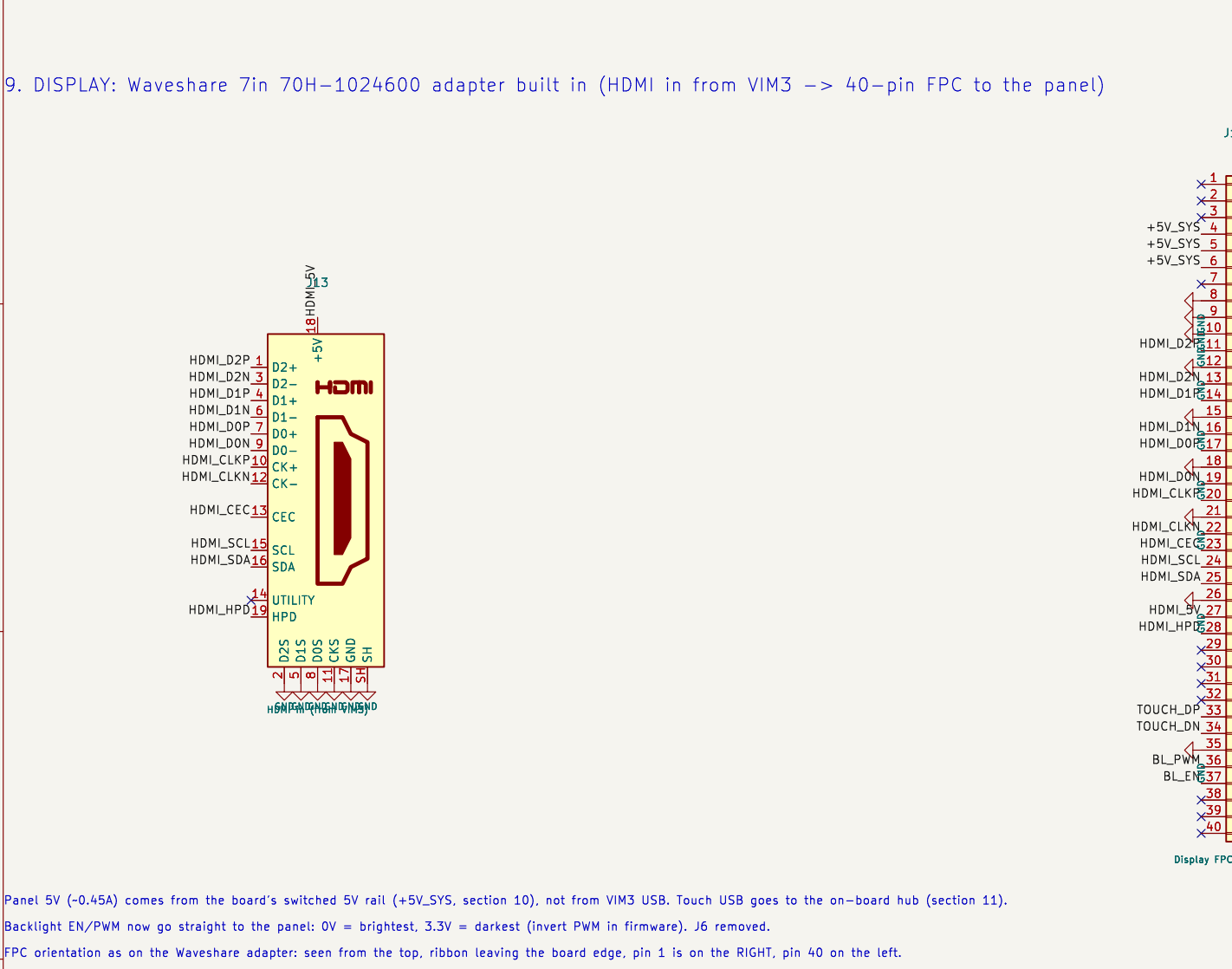
5. GNSS: u-blox NEO-M9N, active antenna on SMA



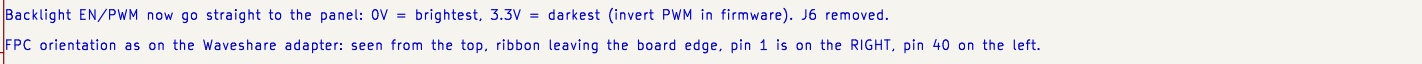
7. AUDIO: VM3 TDM-B I2S -> 2x PCM5102A (front + rear) -> line out on ISO C1



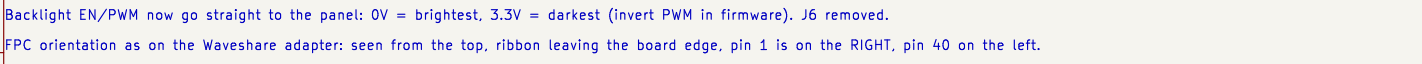
8. DISPLAY: Waveshare Tin 70H-1024x600 adapter built in (HDMI in from VM3 -> A0-pin FPC to the panel)



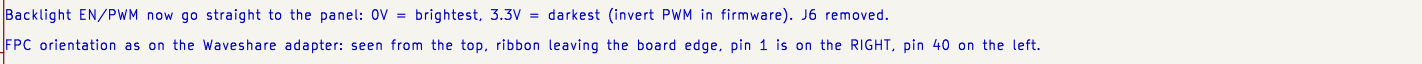
9. DISPLAY: Waveshare Tin 70H-1024x600 adapter built in (HDMI in from VM3 -> A0-pin FPC to the panel)



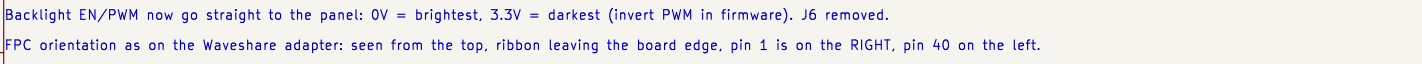
10. 5V/5A SYSTEM SUPPLY FOR VM3 + SCREEN + HUB (TP555288 buck-boost; rides through cranking and overvoltage)



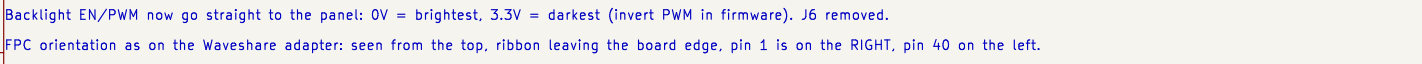
11. MCU CORE (MCU core is section 13)



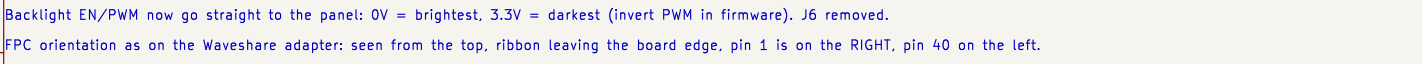
12. EXTERNAL USB PORTS (phone etc.); second hub + dedicated 5V/3A supply, 1.5A current-limited per port



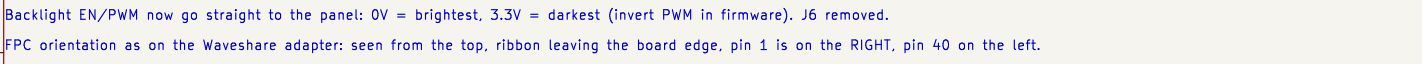
13. MCU: RP2350B (QFN-80) + 16MB QSPI flash + 12MHz crystal + 3.3V LDO, per the Raspberry Pi Pico 2 reference design



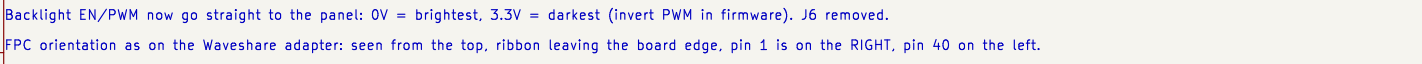
14. USB HUB: VM3 header USB (pins 3/4) -> MCU, screen touch, spare port



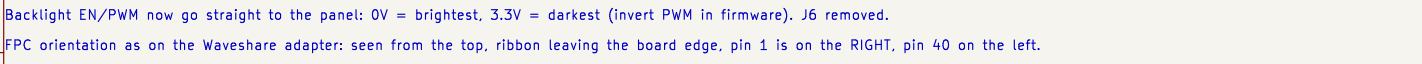
15. VM3 POWER KEY, BACKLIGHT, LIGHT SENSOR (all from the MCU)



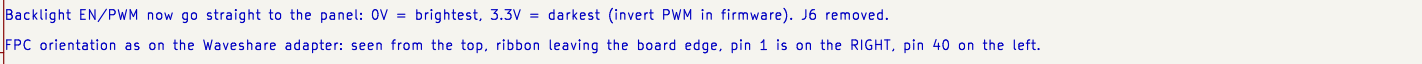
16. VM3 40-PIN HEADER + ISO 10487 C1 (line out)



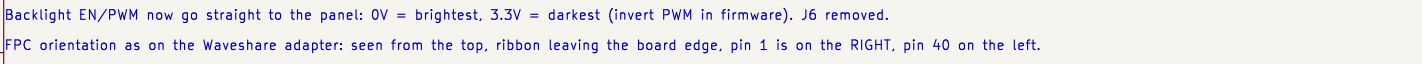
17. VM3 10-pin via 2x20 DCC ribbon



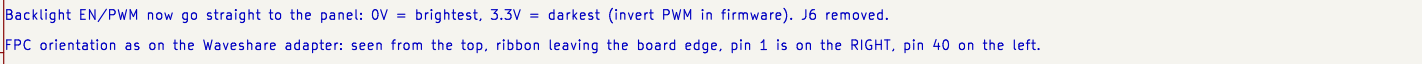
18. VM3 10-pin via 2x20 DCC ribbon



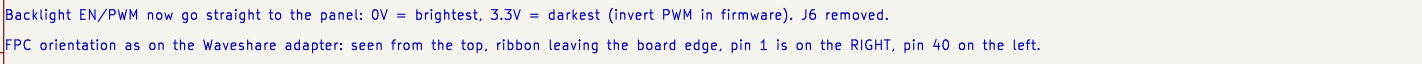
19. VM3 10-pin via 2x20 DCC ribbon



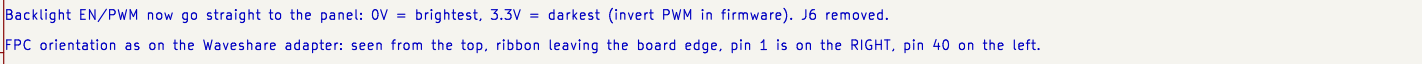
20. VM3 10-pin via 2x20 DCC ribbon



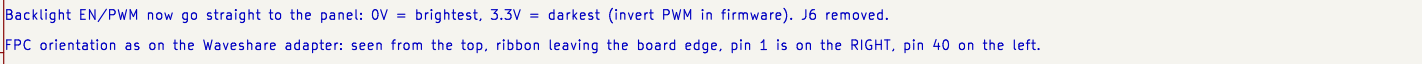
21. VM3 10-pin via 2x20 DCC ribbon



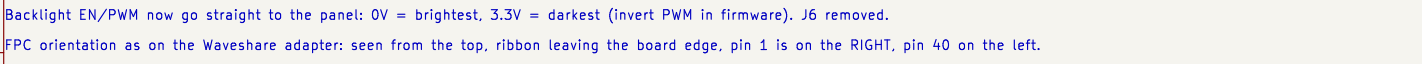
22. VM3 10-pin via 2x20 DCC ribbon



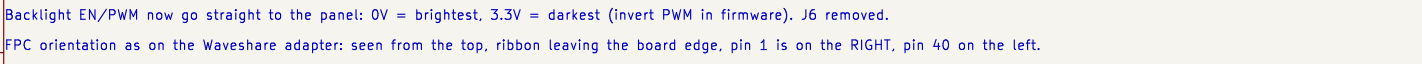
23. VM3 10-pin via 2x20 DCC ribbon



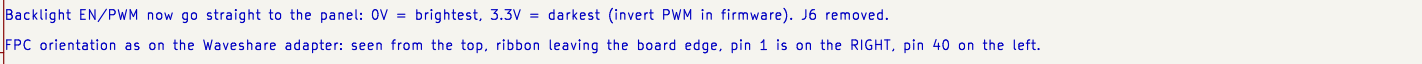
24. VM3 10-pin via 2x20 DCC ribbon



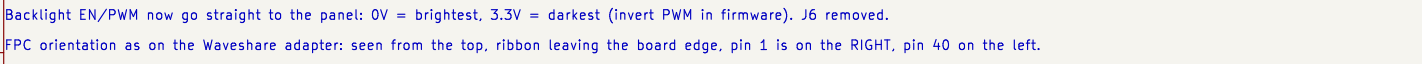
25. VM3 10-pin via 2x20 DCC ribbon



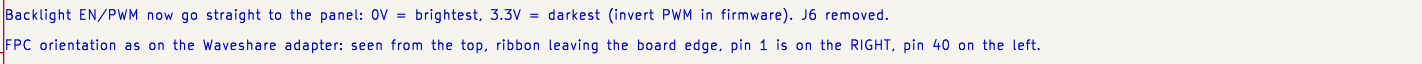
26. VM3 10-pin via 2x20 DCC ribbon



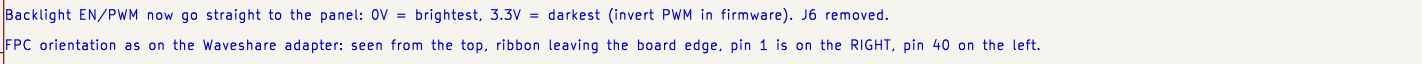
27. VM3 10-pin via 2x20 DCC ribbon



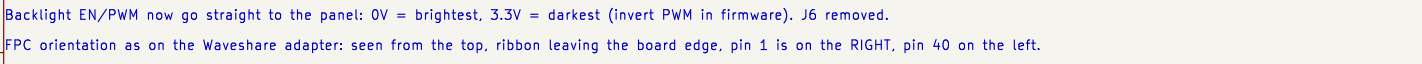
28. VM3 10-pin via 2x20 DCC ribbon



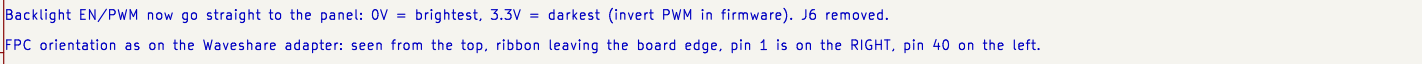
29. VM3 10-pin via 2x20 DCC ribbon



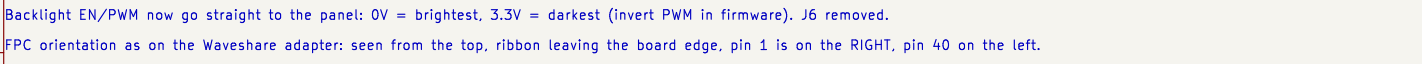
30. VM3 10-pin via 2x20 DCC ribbon



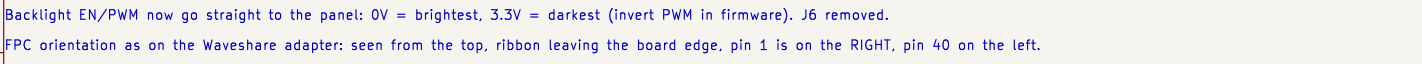
31. VM3 10-pin via 2x20 DCC ribbon



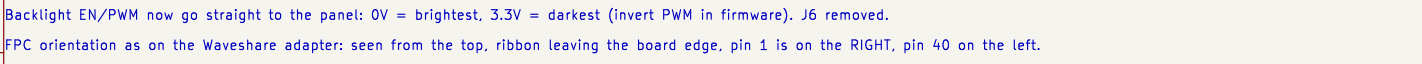
32. VM3 10-pin via 2x20 DCC ribbon



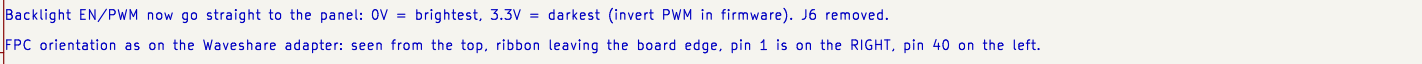
33. VM3 10-pin via 2x20 DCC ribbon



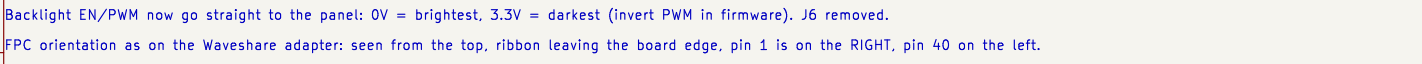
34. VM3 10-pin via 2x20 DCC ribbon



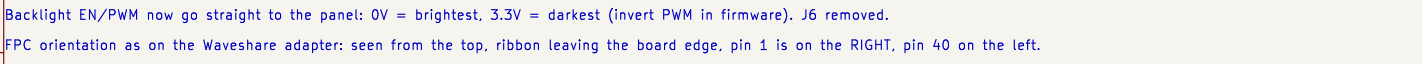
35. VM3 10-pin via 2x20 DCC ribbon



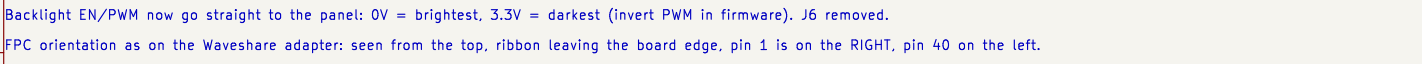
36. VM3 10-pin via 2x20 DCC ribbon



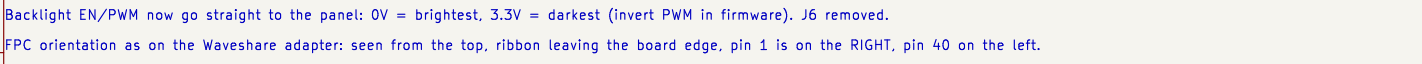
37. VM3 10-pin via 2x20 DCC ribbon



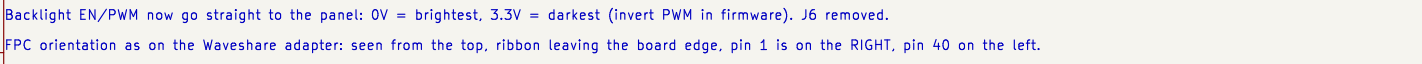
38. VM3 10-pin via 2x20 DCC ribbon



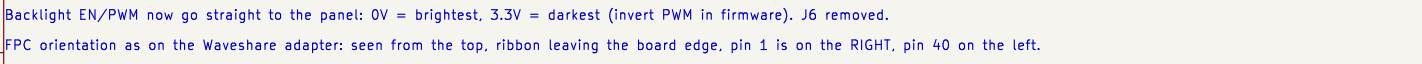
39. VM3 10-pin via 2x20 DCC ribbon



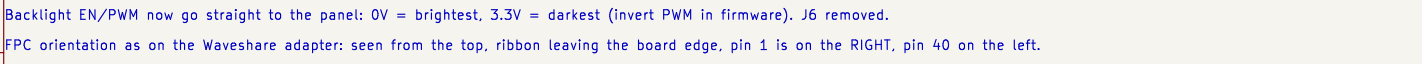
40. VM3 10-pin via 2x20 DCC ribbon



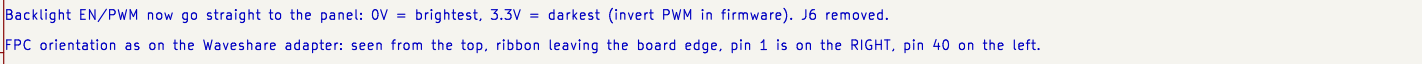
41. VM3 10-pin via 2x20 DCC ribbon



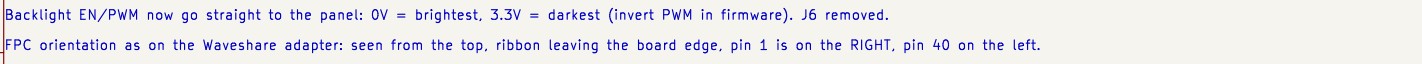
42. VM3 10-pin via 2x20 DCC ribbon



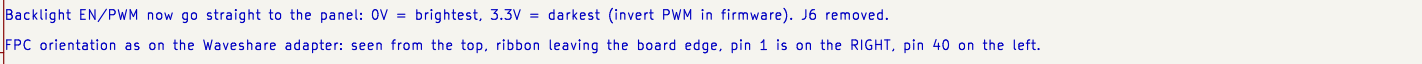
43. VM3 10-pin via 2x20 DCC ribbon



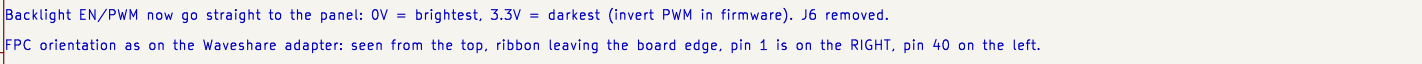
44. VM3 10-pin via 2x20 DCC ribbon



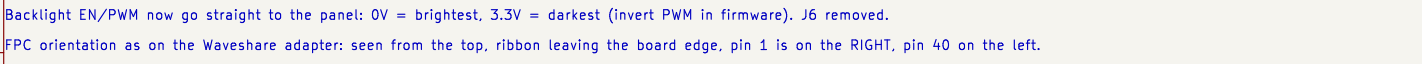
45. VM3 10-pin via 2x20 DCC ribbon



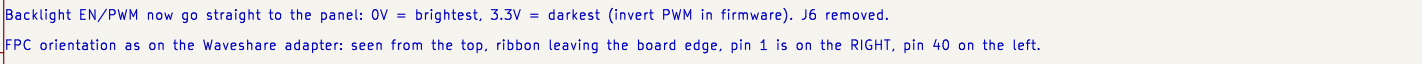
46. VM3 10-pin via 2x20 DCC ribbon



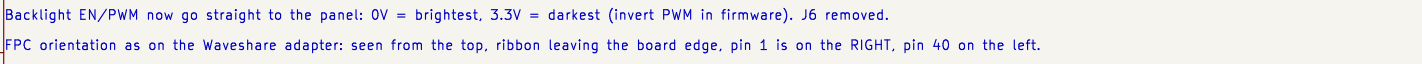
47. VM3 10-pin via 2x20 DCC ribbon



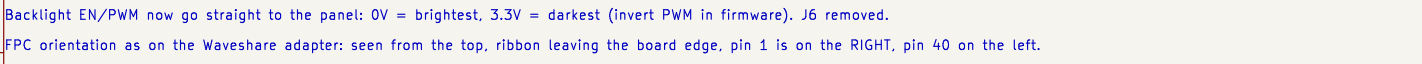
48. VM3 10-pin via 2x20 DCC ribbon



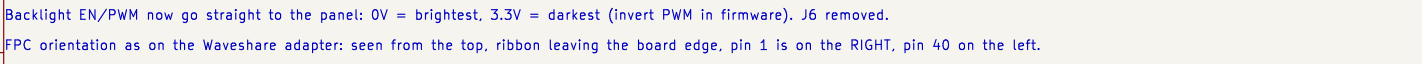
49. VM3 10-pin via 2x20 DCC ribbon



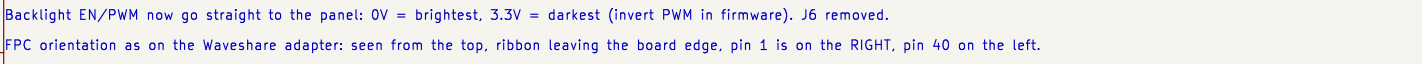
50. VM3 10-pin via 2x20 DCC ribbon



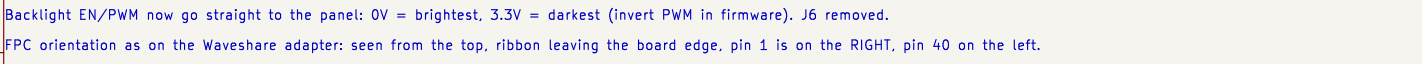
51. VM3 10-pin via 2x20 DCC ribbon



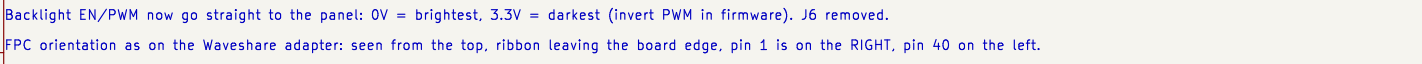
52. VM3 10-pin via 2x20 DCC ribbon



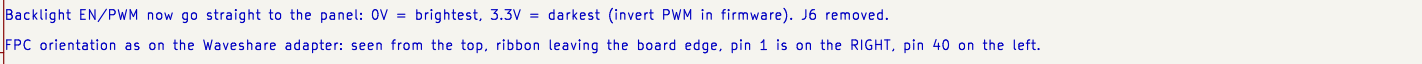
53. VM3 10-pin via 2x20 DCC ribbon



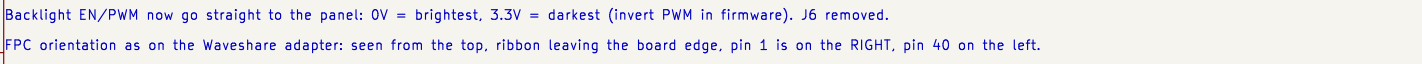
54. VM3 10-pin via 2x20 DCC ribbon



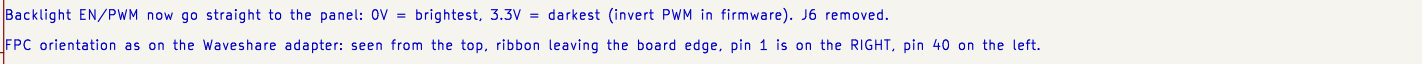
55. VM3 10-pin via 2x20 DCC ribbon



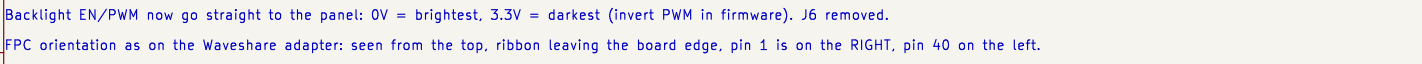
56. VM3 10-pin via 2x20 DCC ribbon



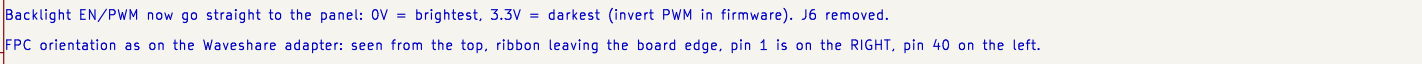
57. VM3 10-pin via 2x20 DCC ribbon



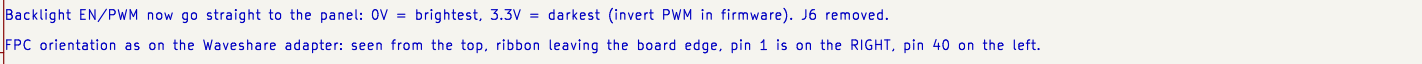
58. VM3 10-pin via 2x20 DCC ribbon



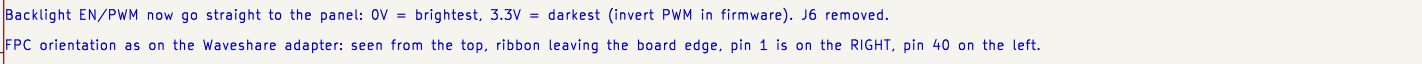
59. VM3 10-pin via 2x20 DCC ribbon



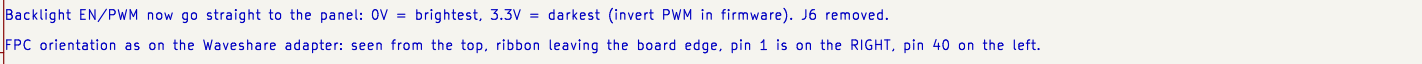
60. VM3 10-pin via 2x20 DCC ribbon



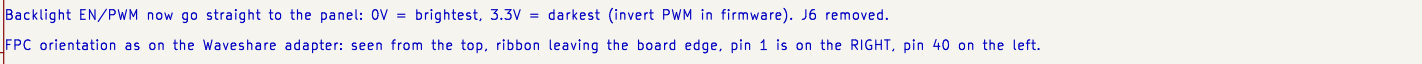
61. VM3 10-pin via 2x20 DCC ribbon



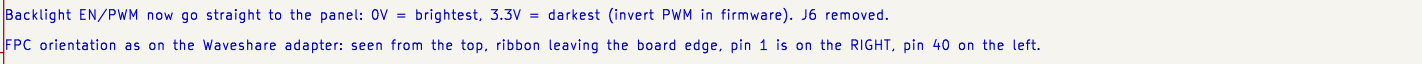
62. VM3 10-pin via 2x20 DCC ribbon



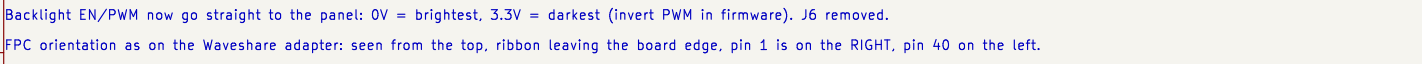
63. VM3 10-pin via 2x20 DCC ribbon



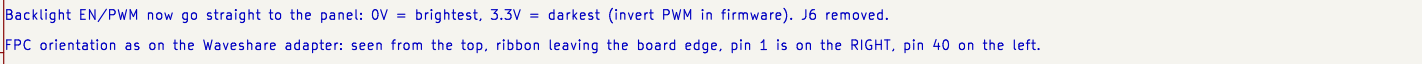
64. VM3 10-pin via 2x20 DCC ribbon



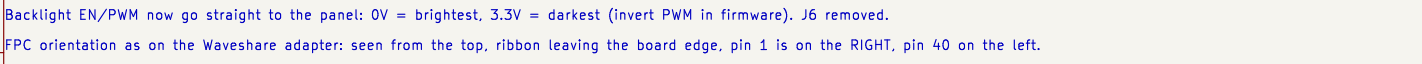
65. VM3 10-pin via 2x20 DCC ribbon



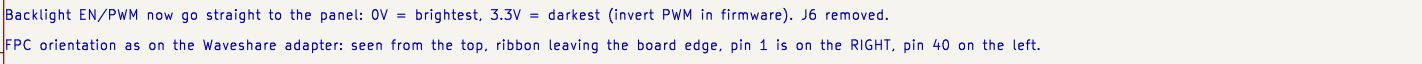
66. VM3 10-pin via 2x20 DCC ribbon



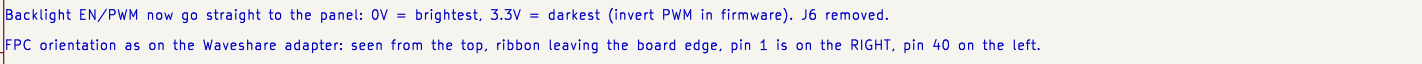
67. VM3 10-pin via 2x20 DCC ribbon



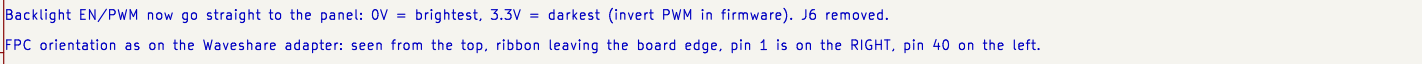
68. VM3 10-pin via 2x20 DCC ribbon



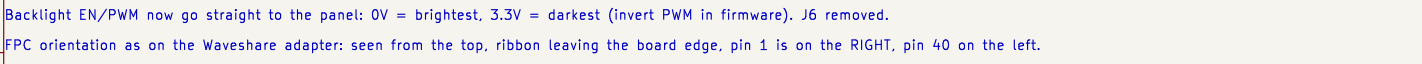
69. VM3 10-pin via 2x20 DCC ribbon



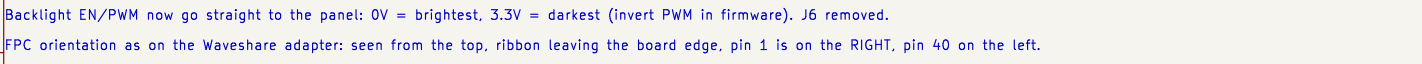
70. VM3 10-pin via 2x20 DCC ribbon



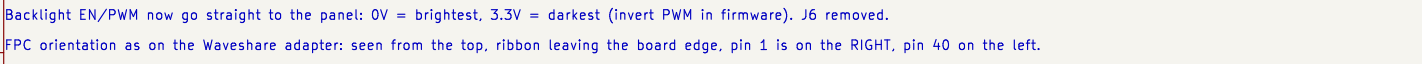
71. VM3 10-pin via 2x20 DCC ribbon



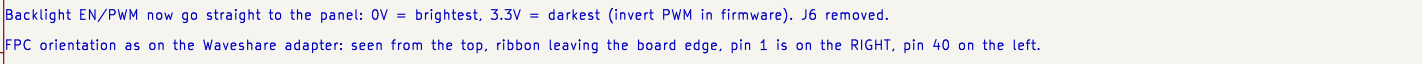
72. VM3 10-pin via 2x20 DCC ribbon



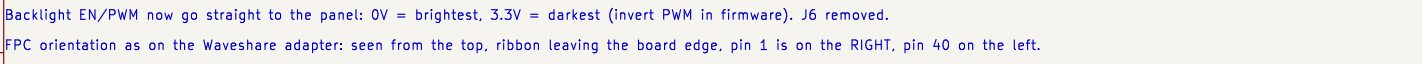
73. VM3 10-pin via 2x20 DCC ribbon



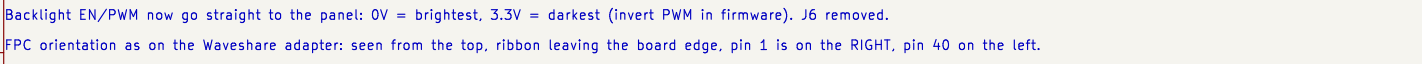
74. VM3 10-pin via 2x20 DCC ribbon



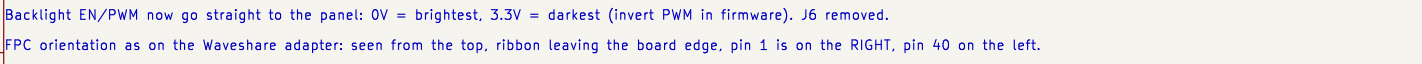
75. VM3 10-pin via 2x20 DCC ribbon



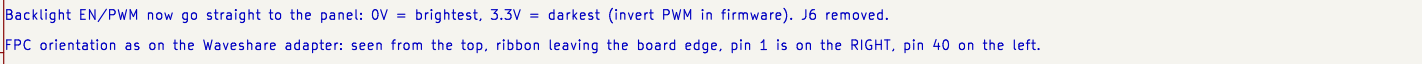
76. VM3 10-pin via 2x20 DCC ribbon



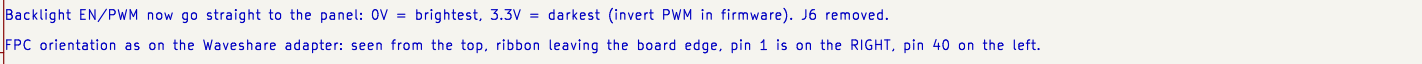
77. VM3 10-pin via 2x20 DCC ribbon



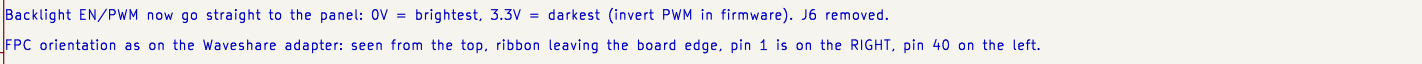
78. VM3 10-pin via 2x20 DCC ribbon



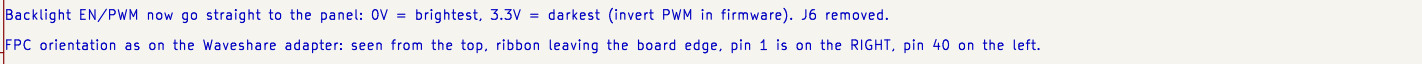
79. VM3 10-pin via 2x20 DCC ribbon



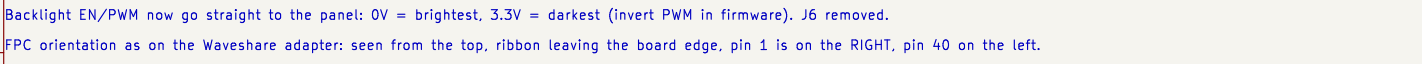
80. VM3 10-pin via 2x20 DCC ribbon



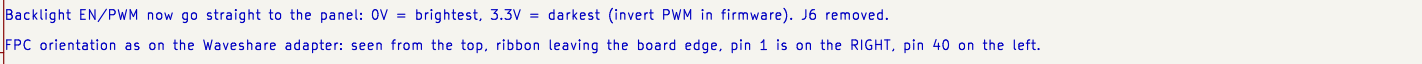
81. VM3 10-pin via 2x20 DCC ribbon



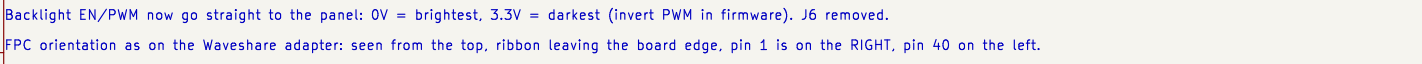
82. VM3 10-pin via 2x20 DCC ribbon



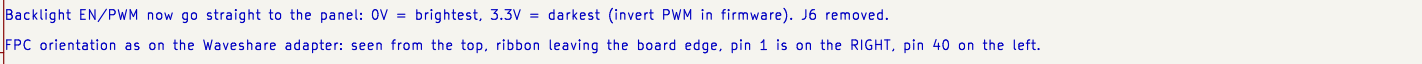
83. VM3 10-pin via 2x20 DCC ribbon



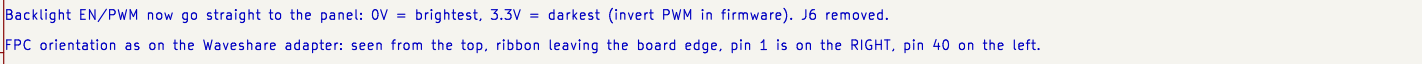
84. VM3 10-pin via 2x20 DCC ribbon



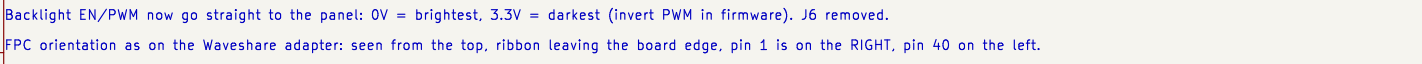
85. VM3 10-pin via 2x20 DCC ribbon



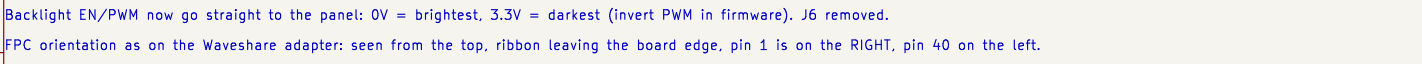
86. VM3 10-pin via 2x20 DCC ribbon



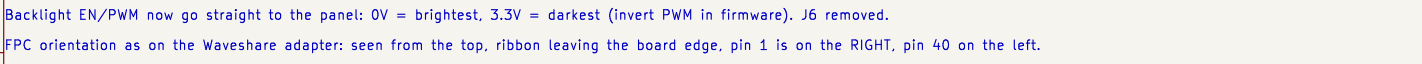
87. VM3 10-pin via 2x20 DCC ribbon



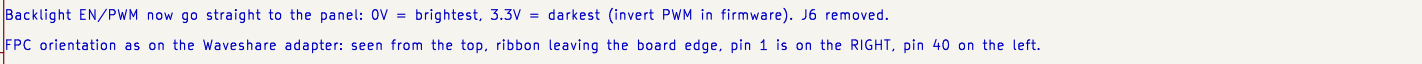
88. VM3 10-pin via 2x20 DCC ribbon



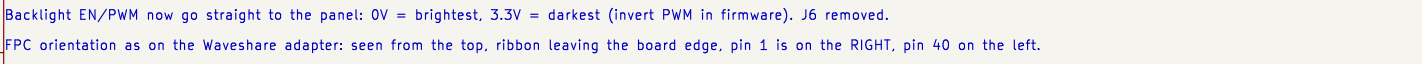
89. VM3 10-pin via 2x20 DCC ribbon



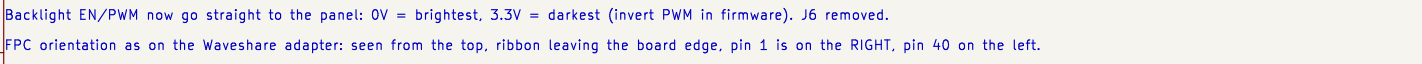
90. VM3 10-pin via 2x20 DCC ribbon



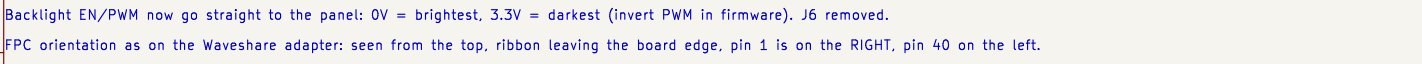
91. VM3 10-pin via 2x20 DCC ribbon



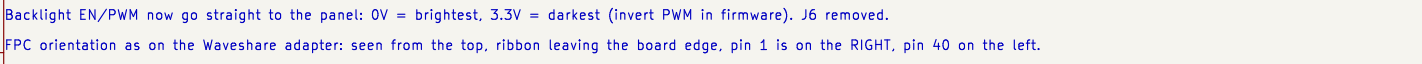
92. VM3 10-pin via 2x20 DCC ribbon



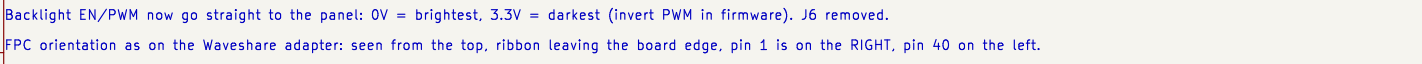
93. VM3 10-pin via 2x20 DCC ribbon



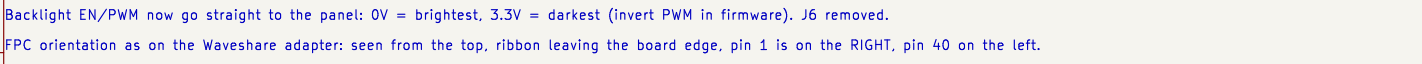
94. VM3 10-pin via 2x20 DCC ribbon



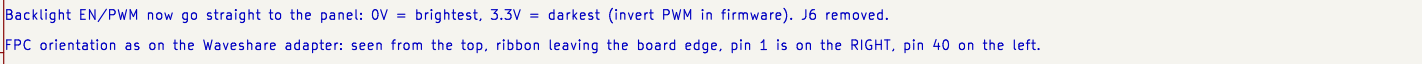
95. VM3 10-pin via 2x20 DCC ribbon



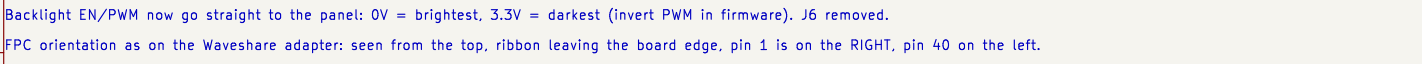
96. VM3 10-pin via 2x20 DCC ribbon



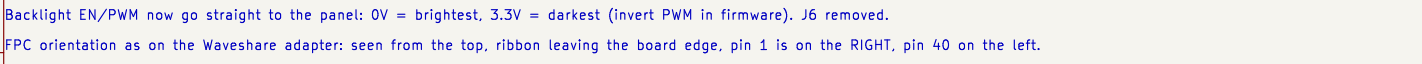
97. VM3 10-pin via 2x20 DCC ribbon



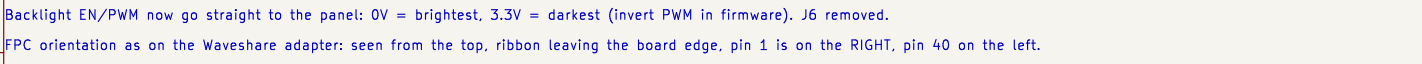
98. VM3 10-pin via 2x20 DCC ribbon



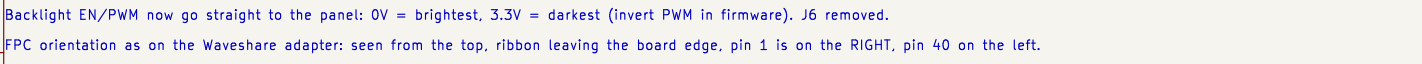
99. VM3 10-pin via 2x20 DCC ribbon



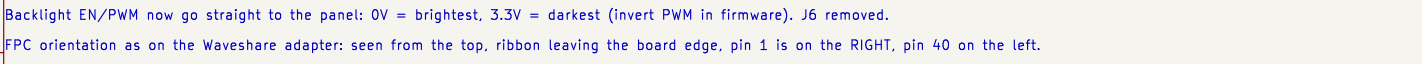
100. VM3 10-pin via 2x20 DCC ribbon



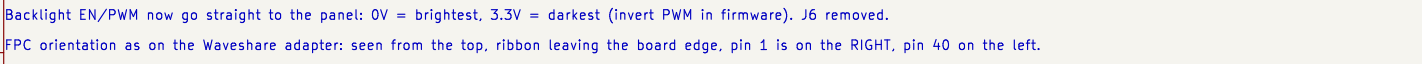
101. VM3 10-pin via 2x20 DCC ribbon



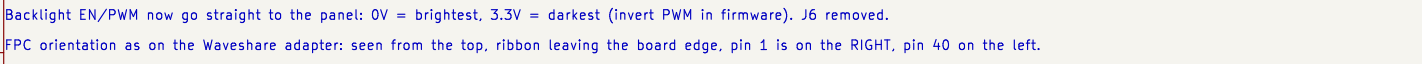
102. VM3 10-pin via 2x20 DCC ribbon



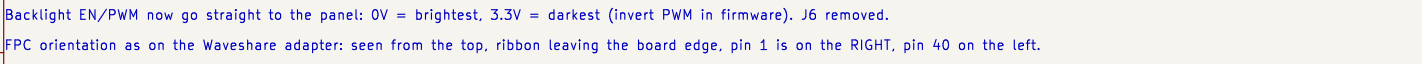
103. VM3 10-pin via 2x20 DCC ribbon



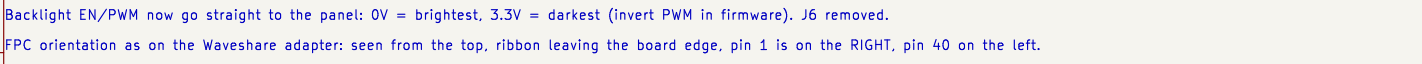
104. VM3 10-pin via 2x20 DCC ribbon



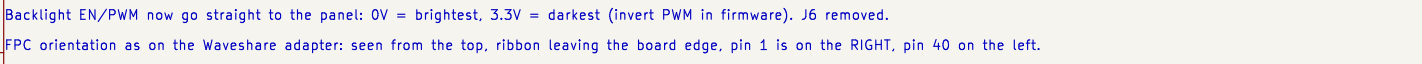
105. VM3 10-pin via 2x20 DCC ribbon



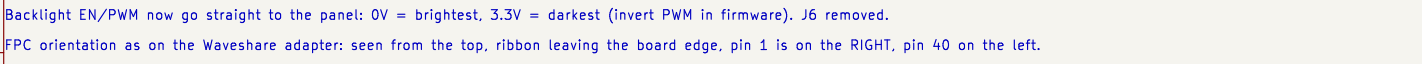
106. VM3 10-pin via 2x20 DCC ribbon



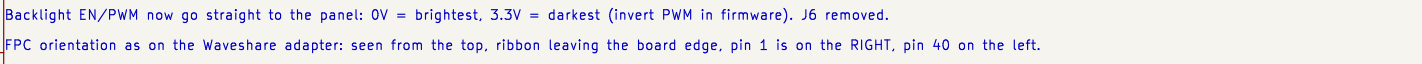
107. VM3 10-pin via 2x20 DCC ribbon



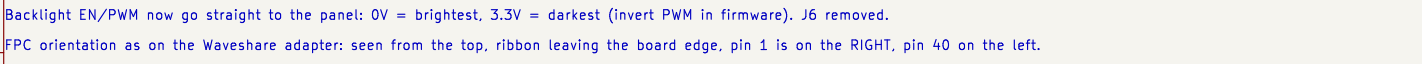
108. VM3 10-pin via 2x20 DCC ribbon



109. VM3 10-pin via 2x20 DCC ribbon



110. VM3 10-pin via 2x20 DCC ribbon



111. VM3 10-pin via 2x20 DCC ribbon

